



THE NATIONAL INSTITUTE OF ENGINEERING

(An Autonomous Institution under VTU, Belagavi)

Manandavadi Road, Mysuru – 570 008.

Department of CSE AND AIML

Course Plan for the term ODD/EVEN 2026-27

Course Instructor: Dr. S Lokesh
Course Title: Digital Design and Computer Organization
LTP: 42:0:28:50

Term: 10/08/2026 to 28/11/2026
Course Code: 1BCS302
Semester: 3 CSE B

LESSON PLAN

Session No.	Planned Date	Content	Actual Date/s	Mode of Delivery (Classroom /Online)	Link for online content	Link for Course Content	Comments, if any by CI/HoD
1.	10-08-2026	Module 1 – Simplification of Boolean Functions: The K-Map Method, Two-and Three Variable Maps,		Chalk & Talk	https://youtu.be/SyNP83xv9IM?si=BUNv47I91POZq13-	https://moodlegurukul.nie.ac.in/mod/folder/view.php?id=91425	
2.	13-08-2026	Four- Variable Maps,		Chalk & Talk + Problem Solving	https://youtu.be/v08rGyVYalI?si=pvr_Ro2Ex2dTEqEd		
3.	14-08-2026	Product of sums simplification, Implementation using NAND and NOR Universal gates, Don't-Care Conditions		Chalk & Talk + Demonstration	https://www.youtube.com/watch?v=w2hK0JV_KmJc		

					https://youtu.be/ZayoUTi2tsA?si=WWJ2sVSTUnd2F6xr		
4.	17-08-2026	Combinational Logic: Design procedure, Half Adder, Full Adder, Half Subtractor, Full Subtractor		Chalk & Talk + Demonstration	https://youtu.be/85XxQZqBNlg		
5.	20-08-2026	EX-OR and logic function realization		Chalk & Talk	https://youtu.be/rfaKzj5go-0?si=-PaNmuPTBmhyUm70		
6.	21-08-2026	Binary Adder		Chalk & Talk + Demonstration	https://youtu.be/tI27Er4h1Yw?si=7L-LjGGOQHwyczEz		
7.	24-08-2026	Decimal Adder		Chalk & Talk + Problem Solving	https://youtu.be/9O7OJi7RCLQ?si=5a7P0928iAU9sKtC		
8.	27-08-2026	Encoders, Decoders,		PPT + Chalk & Talk	https://youtu.be/fBvhLFQEDk?si=4E2-RYzcU3fEN7O-		
9.	28-08-2026	Multiplexers, Demultiplexers		Chalk & Talk + Demonstration	https://youtu.be/HIeQhZ9Gq5s?si=saukAJU52p_cjJmu		

10.	31-08-2026	Module 2-Sequential Circuits Sequential Logic: Introduction, - Latches and Flip-Flops, Triggering of Flip Flops		PPT + Chalk & Talk	https://youtu.be/jm0PGDSSBkI https://youtu.be/jm0PGDSSBkI	https://moodlegurukul.nie.ac.in/mod/folder/view.php?id=91429	
11.	03-09-2026	Flip-Flop excitation tables		Chalk & Talk + Problem Solving	https://youtu.be/k7oc4FLba3w?si=D8jJL78_e8w53Inu		
12.	04-09-2026	characteristic equations		Chalk & Talk	https://youtu.be/k7oc4FLba3w?si=D8jJL78_e8w53Inu		
13.	07-09-2026	Design Procedure with Flip-Flops, Race around condition		Chalk & Talk + Demonstration	https://youtu.be/rPGhO7MPnw?si=UO99EDIEY2-xTvNa		
14.	10-09-2026	Registers: Registers with parallel load, sequential logic implementation		Chalk & Talk	https://youtu.be/q3KYk1iQcEA?si=L1uQYpKejLI_bRH		
15.	11-09-2026	Shift Registers: serial transfer, Bidirectional shift registers with parallel load		Demonstration + Chalk & Talk	https://youtu.be/whUk4TTGj8M?si=hnPSbN33-duM4QYu		
16.	17-09-2026	Synchronous counters		Chalk & Talk + Problem Solving	https://youtu.be/dIianWwz-Fw?si=1TlbLCOIh4JEOAVK		
17.	18-09-2026	MOD-N Design		Chalk & Talk + Demonstration	https://youtu.be/ZTZAYnT9Iaw?si=mPJYRymU-ceuOeaV		

18.	21-09-2026	Module – 3: Processing Unit Introduction: Functional Units, Basic Operational Concepts, Performance		PPT + Flipped Classroom	https://www.youtube.com/watch?v=x59-56NpljE	https://moodlegurukul.nie.ac.in/mod/folder/view.php?id=91428	
19.	28-09-2026	Machine Instructions and Programs: Memory Location and Addresses- Byte addressability, Big Endian and Little Endian Assignment, memory operations,		Chalk & Talk + PPT	https://www.youtube.com/watch?v=xGlVPh2xbhw		
20.	01-10-2026	Instructions and instruction sequencing: Register transfer notation, Assembly Language Notation		Chalk & Talk	https://www.youtube.com/watch?v=op5HGBH62V4		
21.	05-10-2026	Basic Instruction Types, Addressing Modes, Pipeline.		PPT + Chalk & Talk	https://www.youtube.com/watch?v=h-iHq13WhmQ https://www.youtube.com/watch?v=uDXmrMxNNys		
22.	08-10-2026	Execution of complete instruction		Demonstration + Problem-Based Learning	https://www.youtube.com/watch?v=SFend4mLdzY		
23.	09-10-2026	Multiple-Bus organization		Chalk & Talk + Demonstration	https://www.youtube.com/watch?v=ZlAoj1eoLMg		
24.	12-10-2026	Hardwired control unit		Chalk & Talk	https://www.youtube.com/watch?v=yqbsNiNF3Ts		
25.	15-10-2026	Micro programmed control		PPT + Chalk & Talk	https://www.youtube.com/watch?v=qWzYKC5t77o		

26	16-10-2026	Module – 4 : Arithmetic Unit: Multiplication of Positive numbers		Chalk & Talk + Problem Solving	https://www.youtube.com/watch?v=mJe_vm-LcEY	https://moodlegurukul.nie.ac.in/mod/folder/view.php?id=91427	
27.	19-10-2026	A signed operand multiplication		Chalk & Talk	https://www.youtube.com/watch?v=LN-CY0IEgZ4		
28.	22-10-2026	Bit pair recoding of multipliers		Chalk & Talk + Demonstration	https://youtu.be/iDZfHLdiJho?si=f0Z-dX6kL-JLqw3Q		
29.	23-10-2026	Bit pair recoding of multipliers		Problem-Based Learning + Chalk & Talk	https://youtu.be/iDZfHLdiJho?si=f0Z-dX6kL-JLqw3Q		
30.	26-10-2026	carry save addition of summands		Chalk & Talk + Problem Solving	https://youtu.be/YiuUINlgX2g?si=UNwq1GcEOQXMwMts		
31.	29-10-2026	Introduction to Integer division		Chalk & Talk	https://youtu.be/6ToR6vuRb3M?si=zuyroTu5QripLVZm		
32.	30-10-2026	Restoring Algorithm		Chalk & Talk + Problem Solving	https://youtu.be/Ek-Ea2NnjeQ?si=Z2kZCpeK0GufJSXu		
33.	02-11-2026	Non-Restoring Algorithm		Chalk & Talk + Demonstration	https://youtu.be/je7RSR6w37c?si=BZJgF73PF21x-HBc		

34.	05-11-2026	Module 5-Memory Unit Memory Unit: Basic concepts,		PPT + Chalk & Talk	https://www.youtube.com/watch?v=ZeZ_spCH8fM	https://moodlegurukul.nie.ac.in/mod/folder/view.php?id=91426	
35.	06-11-2026	Internal organization of memory chips		Chalk & Talk + Demonstration	https://www.youtube.com/watch?v=_SXrCv9voh8		
36.	09-11-2026	Internal organization of memory chips		Demonstration + Problem-Based Learning	https://www.youtube.com/watch?v=_SXrCv9voh8		
37.	12-11-2026	Structure of larger memory.		Chalk & Talk	https://www.youtube.com/watch?v=h5jvsnpzhz4		
38.	13-11-2026	Structure of larger memory.		PPT + Demonstration	https://www.youtube.com/watch?v=h5jvsnpzhz4		
39.	19-11-2026	Cache memories,		Flipped Classroom + Discussion	https://www.youtube.com/watch?v=AceWN-qQqDk		
40.	20-11-2026	Mapping functions		Chalk & Talk + Problem Solving	https://www.youtube.com/watch?v=fMaqtKEHUT4		
41.	23-11-2026	Cache Policies		Flipped Classroom + Discussion	https://www.youtube.com/watch?v=wfVy85Dqiyc		
42.	26-11-2026	Input Output : Accessing I/O Devices		PPT + Flipped Classroom	https://www.youtube.com/watch?v=Hr7tbmIr0G8		

Laboratory Session							
Session No.	Planned Date/s B1 Batch	Content	Actual Date/s for B1, B2, B3 and B4	Mode of Delivery (Classroom /Lab/Online/Virtual)	Link for online content/Virtual lab	Link for Course Content	Comments, if any by CI/HoD
1.	10/08/2026	Realize Basic gates and Universal gates and Verify it's truth table		LAB	Electronics Lab experiment-1 : Realization of NOT, AND, OR & X-OR gates using NAND gates (IC-7400)	https://moodlegurukul.nie.ac.in/mod/folder/view.php?id=91442	
2.	17/08/2026	Implement Half adder and Full adder using logic gates. Implement Half subtractor, Full subtractor using logic gates		LAB	https://youtu.be/pFiFayVTXgA?si=ziZlbe0uxqA2pMOO		
3.	24/08/2026	Implement BCD to Excess-3 using basic gates		LAB	https://youtu.be/Hy1lxxJAlgo?si=ko9IlpLiRj2OnLR1		
4.	07/09/2026	Realize a full adder or any Boolean function using 3:8 decoder IC.		LAB	https://youtu.be/wiwFmoodqks?si=Y7C5xB_nj-qiVKG9		
5.	21/09/2026	Given any four variable logic expression, realize the simplified logic expression using 8:1		LAB	https://youtu.be/QzgGqIT5M0U?si=o5skWRokn-SJxRkw https://youtu.be/INbbqSVjDHc?si=1kEXJYGrxsJEhoR		
6.	28/09/2026	Realize SR, J-K, T and D flip flops and verify their truth tables Realize SR, J-K, T and D flip flops and verify their truth tables		LAB	Electronics Lab experiment-4 : Realization of		

					SR flip-flop using NAND gates (IC-7400) Electronics Lab experiment-5 : Realization of JK & T flip-flop IC-7476 D Flip flop experiment		
7.	05/10/2026	Realize a JK master Slave Flip- Flop and verify its truth table.		LAB	https://youtu.be/3Xudu0rIKuU?si=wRVMR3EX7hW2nC4c		
8.	12/10/2026	Realize mod-4 and mod-6 counter using Synchronous design		LAB	https://youtu.be/yAcfRp2jDSw?si=WRZSlfK5nbPyrzEP		
9.	19/10/2026	Design Verilog HDL to implement simple circuits using structural, Data flow and Behavioural model.		LAB	https://youtu.be/Pl_b5V4leSU?si=2uxa_j6BvC6Q1Wbh		
10	26/10/2026	Write a VHDL code for a Half –Adder Simulate and verify its working.		LAB	https://youtu.be/ZrsH2L6VBYo?si=PQ0UQREuZnKCOu3u		
11	02/11/2026	Write a VHDL code for a Full –Adder , Simulate and verify its working.		LAB	https://youtu.be/U1NTVhtW5ms?si=1fl-wHeBD7WUzch5		
12	09/11/2026	Write a VHDL code for a Full –Subtractor, Simulate and verify its working..		LAB	https://youtu.be/W_oQPxK9zlM?si=TSa6JjnbmKxYHRld		
13	23/11/2026	Write a VHDL code for 2:1, 4:1 and 8:1 multiplexer , Simulate and verify its working.		LAB	https://youtu.be/rFASpGnNrg?si=C70H9B5vBswj		

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14	25/11/2026	Write a VHDL code for a SR, D and JK flip flop , Simulate and verify its working.		LAB	https://youtu.be/a9j4EjIT5R0?si=nTv6r8scEoIF8xPz		

Evaluation pattern

BT Level	Test-1 (25 Marks)	Test-2 (25 Marks)	Quiz (20 Marks)	Experiential Learning	SEE (100 Marks)
L1	-	-	-	-	-
L2	05	05	06	-	20
L3	10	10	10	10	40
L4	05	05	04	10	30
L5	05	05	-	10	10

Experiential Learning and Evaluation

Component	Sub-Component	Evaluation Criteria	Marks
Lab Continuous Internal Evaluation (Lab CIE): 10 marks	Experiment Execution	Correct implementation and completion of experiments	5
	Lab Record	Timely submission and quality of documentation	5

Lab Test: 10 marks	Program/ Experiment Development	Correct logic, coding, or experimental procedure	5
	Output & Result Analysis	Accuracy of results and interpretation	2
	Viva Voce	Understanding of concepts and methodology	3
Anyone Task -10Marks • Research and Theorotical analysis (The Literature Task) • Schematic Capture & System Validation (The simulation/ Logisim Task) • Hardware Implementation & Silicon Verification (The Breadboard Task)	Problem Identification	Understanding and formulation of the problem	2
	Analysis & Solution Design	Application of concepts and analytical skills	4
	Report Preparation	Organization, presentation, and documentation	2
	Presentation & Discussion	Communication and response to questions	2
Suggested Tools/ AI assisted Tools : Logisim, Multisim, Proteus , Tinkercad, Wokwi, Breadboard and Electronic Components			

Signature of Course Instructor

Signature of Course Coordinator/Faculty Mentor

Signature of HoD